Worksheet 10.2

Science Quest 9: pages 351–5

Conductors and insulators

Student: ..... Class: .....

Comprehension

1. Read the following text and answer the questions.

Conductors are materials that readily allow electric charges to flow through them when a voltage is applied across them. In the late 1800s it was discovered that it was the electrons from the atoms making up the conductor that moved through the material and that the protons were inside the nucleus and remained fixed in position.

Metals are good conductors of electrcity and non-metals are not. An exception to this is graphite, a form of carbon that is a conductor. The conductivity of a metal decreases as the temperature of that conductor increases. This is explained by the increase in resistance to the flow of electrons through a conductor as it heats up. Resistance to electron flow is caused by vibrations of the atoms hindering the flow of electrons through a material. As the temperature increases the vibrations increase and so electron flow is reduced.

Non-metals are classified as insulators. When a voltage is applied across the ends of an insulator almost no current flows as the electrons are not free to move through the material. If electrical charges are transferred to one location on an insulator, they remain there and do not move about. Insulators are important as they are used in electric circuits to prevent us making direct contact with current-carrying wires and so preventing electrocution.

(a) Identify the subatomic particle that moves when a voltage is applied across a metal conductor.

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(b) Explain why the conductivity of a metal increases as a metal is cooled.

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(c) Ebonite is an insulator. When it is rubbed with wool it acquires a negative charge.

(i) Where did these charges come from?

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(ii) Will the negative charges move along the ebonite rod if a voltage is applied across its ends? Explain.

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Comparing conductivity

2. The following table compares the conductivity of various metals as well as graphite and silicon to the conductivity of silver, which is given an arbitrary conductivity rating of 10 000 units.

Element	Conductivity
Silver	10 000
Copper	9460
Gold	7143
Aluminium	5984
Zinc	2634
Tin	1456
Graphite	10
Silicon	0.000 000 02

(a) Plot a column graph to compare the conductivity of each of the materials.

(b) Explain why it is more appropriate to present this data using a column graph than a line graph.

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(c) Even though silver and copper provide better conductivity, most high-voltage wires are made of aluminium (and steel). Suggest a reason for this.

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